

Roll No.:

END SEMESTER EXAMINATION JUNE 2025

Name of the Program: B. Tech CSE SPL

Semester: VI

Name of the Course: Network and System Security

Course Code: TCS 679

Time: 3 Hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1	(20 Marks)	
(a)	Explain the different types of security services defined by the OSI Security Architecture. How do these security services contribute to overall network security?	CO2
(b)	How does Multi-Factor Authentication (MFA) strengthen the network access security model? Discuss the role of different authentication factors (something you know, something you have, and something you are) in ensuring secure access.	CO1 CO1
(c)	Describe network security model. How does a network security model guide the design and implementation of secure network infrastructures?	
Q2	(20 Marks)	
(a)	Explain the difference between block ciphers and stream ciphers in symmetric encryption. Provide examples of each and describe their advantages and disadvantages.	CO3
(b)	What are the main types of cryptanalysis methods used in breaking ciphers? What is a ciphertext-only attack? How does it limit the attacker?	CO3
(c)	Describe the working of the Diffie-Hellman Key Exchange Protocol. What are its vulnerabilities, and how can they be mitigated?	CO2
Q3	(20 Marks)	
(a)	Suppose you choose two prime numbers, i.e., $p=5$ and $q=11$. Use the RSA algorithm to generate the public and private keys. Further encrypt the plaintext message $M=9$. Furthermore, decrypt the ciphertext to recover the original message.	CO4
(b)	Define authentication tied to plaintext. How does it enhance the security of encrypted messages?	CO2
(c)	Explain the main components/fields of an X.509 certificate. How do each of these fields contribute to security?	CO5

Q4	(20 Marks)	
(a)	Explain how Single Sign-On (SSO) works in a federated identity management system. How does identity federation enhance user experience in multi-organizational collaborations like universities, healthcare systems, or government services?	CO5
(b)	How does Pretty Good Privacy (PGP) ensure confidentiality, integrity, and authentication simultaneously? Provide a detailed explanation.	CO6
(c)	How does IPSec secure IP communications across networks? Explain the differences between IPSec Transport mode and Tunnel mode. In which scenarios is each used?	CO4
Q5	(20 Marks)	
(a)	Describe the procedure of SSL handshake protocol. What steps are involved before a secure session is established? Use a figure in your answer.	CO3
(b)	Differentiate between network-based firewalls and host-based firewalls. Explain how intrusion prevention capabilities can be integrated into firewalls. What challenges are faced in such integration?	CO6
(c)	What is a botnet? What role do botnets play in large-scale cyber attacks like Distributed Denial-of-Service (DDoS)? How can such attacks be prevented?	CO5